Connected and Uncooperative: The Effects of Homogenous and Exclusive Social Networks on Survey Response Rates and Nonresponse Bias

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Abstract

Social capital, the strength of people's friendship networks and community ties, has been hypothesized as an important determinant of survey participation. Investigating this hypothesis has been difficult given data constraints. In this paper, we provide insights by investigating how response rates and nonresponse bias in the American Community Survey are correlated with county-level social network data from Facebook. We find that areas of the United States where people have more exclusive and homogenous social networks have higher nonresponse bias and lower response rates. These results provide further evidence that the effects of social capital may not be simply a matter of whether people are socially isolated or not, but also what types of social connections people have and the sociodemographic heterogeneity of their social networks.

* Any opinions and conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7531477: CBDRB-FY23-0260, CBDRB-FY23-0452).

1. Introduction

Survey response rates have been declining across the entire survey industry, increasing survey costs and concerns about higher nonresponse bias. There have been multiple proposed hypotheses, such as people receiving too many survey requests (Leeper 2019) or people becoming busier over time (Vercruyssen et al. 2011). One additional hypothesis is that declining social capital is leading to lower response rates, which we will denote as the "social capital hypothesis." In this paper we define social capital as the strength of people's friendship networks and community ties. Other terms referring to the same or similar concept include social integration (Amaya and Harring 2017) or social isolation.

For a given survey at a given point of time, the social capital hypothesis states that more socially connected individuals are more likely to respond to a survey than more socially isolated individuals. Possible mechanisms could be that socially connected individuals feel more social pressure to respond or feel more strongly that their participation will yield benefits for their social group (Amaya and Harring 2017). Survey sponsors often highlight how the results from their survey will benefit someone's community. From the perspective of Leverage-Saliency Theory (Groves, Singer, and Corning 2000), this pitch will likely be more salient (i.e. influential) for individuals with higher social capital. This will cause them to assign more positive leverage (i.e. weight) to the community benefit aspect of the survey, so they will likely respond to the survey even if there are some survey attributes that discourage their participation (e.g. privacy concerns over financial questions).

In the social capital hypothesis, the two key parts are 1) social capital affects survey participation and 2) social capital has been declining over time. For the latter part, there is evidence that social capital has been declining in the United States, the country we focus on for this paper. Buecker et al. (2021) provide evidence from a meta-analysis that loneliness in young adults has indeed been increasing over time. Fry and Parker (2021) note the share of adults living without a spouse or married partner has been increasing since 1990. There has also been documentation of a decrease in the percent of people who say they have any close friends (Cox 2021), who attend a religious service in the past 7 days (Brenan 2021), and who say that most people can be trusted (NORC at the University of Chicago 2023).

There has been some evidence that social capital affects response rates. Groves, Singer, and Corning (2000) in providing evidence for their leverage-salience theory showed that more civically engaged individuals were more likely to respond to a follow-up survey. Abraham et al. (2009) and Van Ingen et al. (2009) find that volunteering is associated with higher survey participation. Brick and Williams (2013) examine nonresponse in multiple surveys over time, and find that rates of violent crime, number of single-person households, and commute time are associated with lower response rates.

However, given the concept of social capital is multifaceted and has various definitions, not all forms of social connections may have the same relationship to survey participation. Amaya and Harring (2017) find that this may indeed be the case. They find that civic engagement was associated with higher survey unit response but having family and neighbor connections did not have a positive impact (for the most part). In fact, for one of their surveys, they found that individuals close to both friends and family had lower survey response rates. The intuition

here is that civic engagement is more related to seeing the social benefits of survey participation. On the other hand, having tighter family or neighbor connections could be associated with having a more insular social network and being leery of interview requests from outsiders.

In this paper, we provide new evidence that the social capital hypothesis is more complicated than an isolated vs not isolated dichotomy. We use the American Community Survey (ACS) to examine geographic variation in response rates and nonresponse bias, as measured by linked administrative data. We then see how these data are related to Facebook social network data constructed by Chetty et al. (2022). Through various analyses, we find that areas with more homogenous and exclusive social networks have lower response rates and higher nonresponse bias.

Our results make several contributions to the literature. We make use of newly available data on people's social networks, allowing us to explore a new measure of social capital and social integration not available before. The large sample size of the American Community Survey allows us to use these data in geographic analyses and still have statistical precision. And finally, we examine how social capital affects nonresponse bias, while prior literature has only examined unit response rates. This is important because bias is the concept practitioners ultimately care about, particularly with the well known weak correlation between response rates and nonresponse bias (Groves and Peytcheva 2008).

In summary, our results suggest that the effect of social capital on unit response is not solely driven by how many social ties are present, but perhaps also whether social networks include

people from a variety of backgrounds. In other words, if people have sociodemographically heterogeneous social networks, they may be more receptive when an interviewer with different characteristics shows up on their doorstep or gives them a phone call or to self-respond to a survey that yields benefits beyond their insular social networks.

2. Data

The ACS is the largest household survey in the United States, sampling approximately 3.5 million addresses each year, divided into twelve monthly panels for continuous interviewing. It is a multimodal survey where most households are given two months to respond by web or mail, followed by in-person follow-up in the third month for a subsample of the panel. The survey contains questions on a variety of social, housing, and economics topics, and is the key source of such statistics at the county or subcounty level in the United States. The uses of this data are limitless, but the ultimate purpose is to aid the allocation of trillions of dollars in federal funding. For this paper, we analyze ACS households sampled in 2021. In this time period, in-person interviewing was occurring for the most part, with some areas of the country being restricted to phone-call only follow-up at the start of 2021 due to COVID-19. Nevertheless, response rates for the ACS in data year 2021 were 85.3%, which is lower than the response rate of 93% and above for most years in the 2000s and 2010s (American Community Survey Office, n.d.).[2]

For examining nonresponse bias, we merge on various administrative data from the Internal Revenue Service (IRS), Social Security Administration (SSA), and a few other sources to ACS

[2] This is a weighted AAPOR RR6 response rate, with weights also accounting for subsampling of nonrespondents for in-person follow-up (AAPOR 2023).

sampled addresses, including both respondents and nonrespondents. This gives us a measure of how nonrespondents differ before any weighting adjustments. A description of the variables we look at and their primary source are given in Table 1. A more detailed discussion of the administrative data and how they are linked are given in Appendix A.1.

Table 1: Administrative Data Variables and Sources

Variable	Source
Income Over $130,000	IRS Forms 1040, 1099, and W-2
Income Under $40,000	IRS Forms 1040, 1099, and W-2
Household Member Over 60	Social Security Administration Numident File
Household Member Under 10	Social Security Administration Numident File
Any Non-Citizen in Household	Social Security Administration Numident File
Any Persons of Color in Household	2010 and 2020 Decennial Census
Household Member Disabled	Social Security Administration Program Benefit Data
Married Couple in Household	Filing Status from IRS Form 1040
Structure Type is a Single-Family Home/Townhouse	Census Bureau Master Address File

Facebook Data

The Facebook data we link in at the county-level come from Chetty et al. (2022). The two primary variables we use are economic connectedness and clustering. The economic connectedness variable represents the percent of friends a low-SES (socioeconomic status) person has who are high-SES. SES is defined by income, and national median income is used to classify someone as low-SES or high-SES. Income is not observed directly for these Facebook users, but rather is estimated by Chetty et al (2022) using race, education, and location. Therefore, while this variable is framed in terms of income, it ultimately reflects the degree to

which people have friends on Facebook who are of a different race and/or education level, and who live in different neighborhoods than themselves.

The clustering variable represents the average fraction of an individual's friend pairs who are also friends with each other. People with higher levels of clustering are perceived to have more insular social networks. Both these variables are constructed for people who are between 25 and 44, who have been active on Facebook once in the past 30 days, and have at least 100 Facebook friends. Thus, our Facebook data measures the social characteristics of younger adults, while the ACS data represents all ages groups. It is important to note that these two key variables measure the characteristics of a person's Facebook friends, not their overall number of friends, and we have no measure of how close any of these friendships are. Thus, we are focusing on two specific dimensions of people's social networks: diversity and insularity.

We use two additional variables from Chetty et al. (2022): proxies for "volunteering" and "civic organization". Their proxy of volunteering is the percentage of Facebook users who are members of a group which is predicted to be about 'volunteering' or 'activism' based on group title and other group characteristics. Their proxy of civic organization is the number of Facebook Pages predicted to be "Public Good" pages based on page title, category, and other page characteristics, per 1,000 users in the county. Note that the volunteering measure is based on Facebook group membership and doesn't necessarily measure how often people volunteer. Similarly, the civic organization measure is based on the number of certain types of Facebook groups and doesn't necessarily measure a community's degree of civic engagement.

Geography

The ACS response rates, ACS bias, and Facebook data are all linked at the county-level. Because small counties have a small sample size in the ACS 1-year, we make use of two levels of aggregation. The first is Core-Based Statistical Areas (CBSA). For counties not a part of any CBSA, these counties are all grouped together within a state to form a group. The second is ACS estimation areas for the 2021 1-year. These are the county groupings used in the ACS weighting process. Counties with large or medium-sized populations are their own areas, and counties with small populations are grouped together within state. From the 3,143 counties and county equivalents in the United States as of 2021, the CBSA approach yields a count of about 1,000 county groups and the ACS estimation area approach yields a count of about 2,100 county groups.[3]

3. Maps and Correlations

To describe how these data are related with each other, we first graph the data. Figure 1 presents housing unit response rates to the 2017-2021 ACS 5-year at the county level. Figures 2-6 present the bias estimates by CBSA group for the 2021 ACS sample year. Figures 7-8 present the two Facebook variables graphed at the county level.

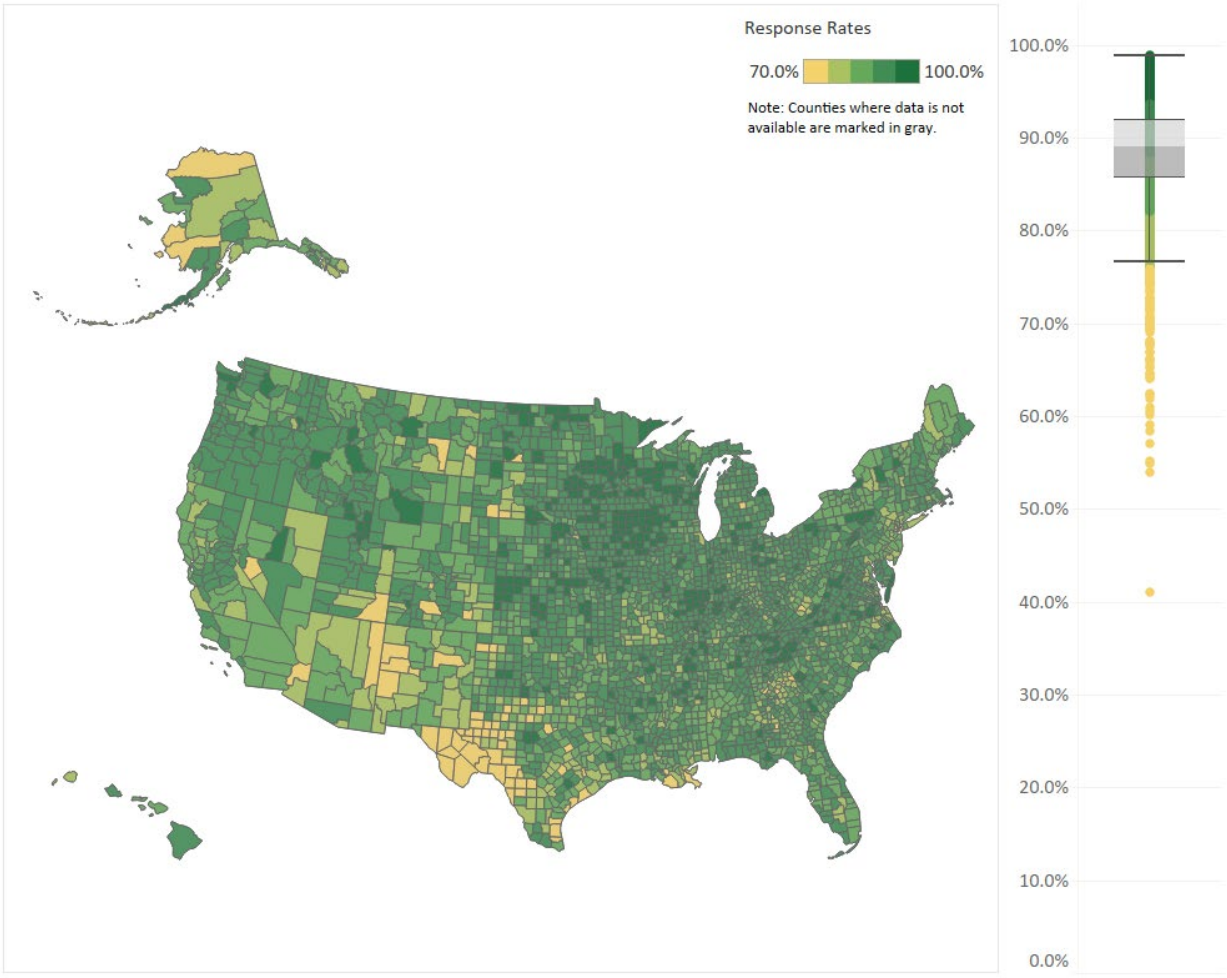
For a linked administrative data variable, Figures 2-6 show the mean for just respondents minus the mean of all occupied housing units (including nonrespondents). In other words, we estimate $\bar{Y}_{v,c}^{R} - \bar{Y}_{v,c}$, the standard definition of nonresponse bias, for each variable v and county group c. In this bias measure, note that these statistics derived from administrative data, not the survey data. So many of these measures are proxies for the variables collected in the ACS. Additionally, the means are weighed means using the survey base weights, so these differences

[3] County group counts are rounded according to Census disclosure rounding rules.

represent the level of bias in the survey before any weighting adjustments (e.g. calibration to Census Population Estimates).[4] We focus on the unadjusted means for respondents since we are primarily interested how the characteristics of respondents differ from nonrespondents, not how well statistical weighting methods perform at removing these differences.

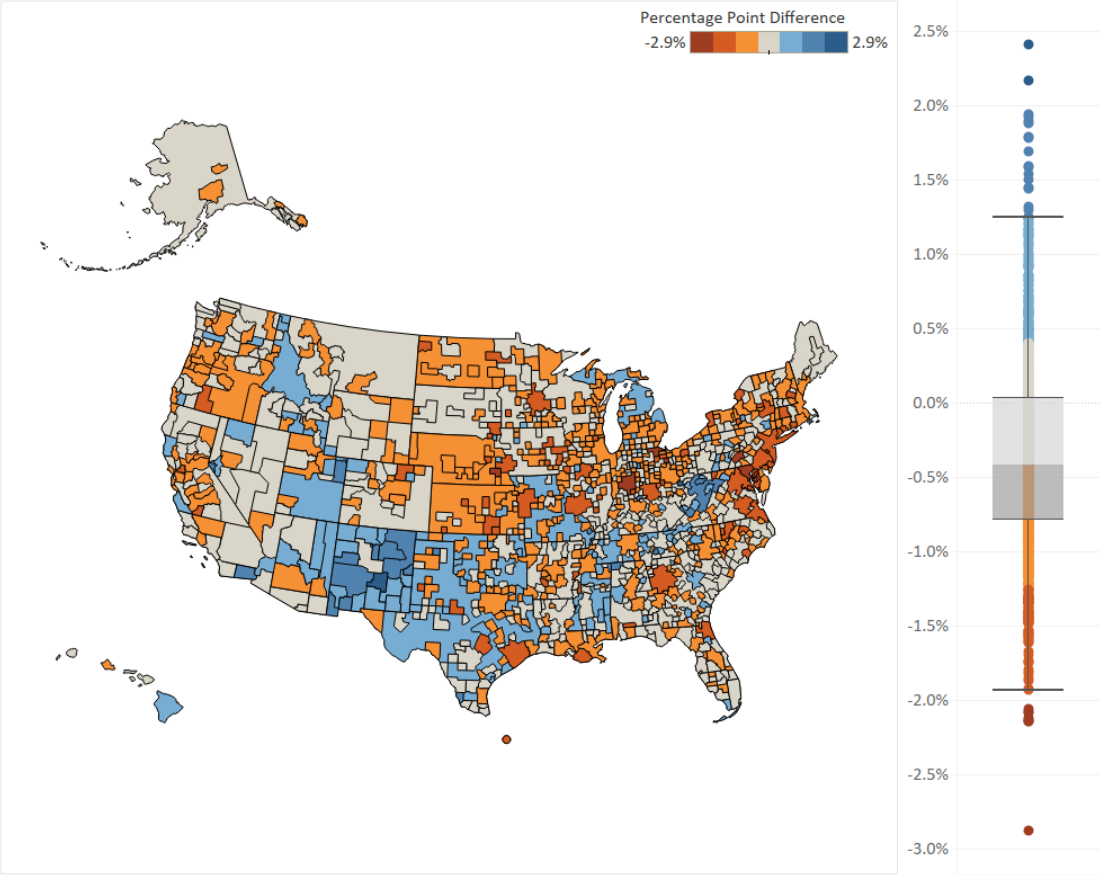
[4] Specifically, the weight used here is the ACS base weight times the Computer-Assisted Personal Interviewing subsampling factor (American Community Survey Office 2022).

Figure 1: ACS 2017-2021 5-year Response Rates



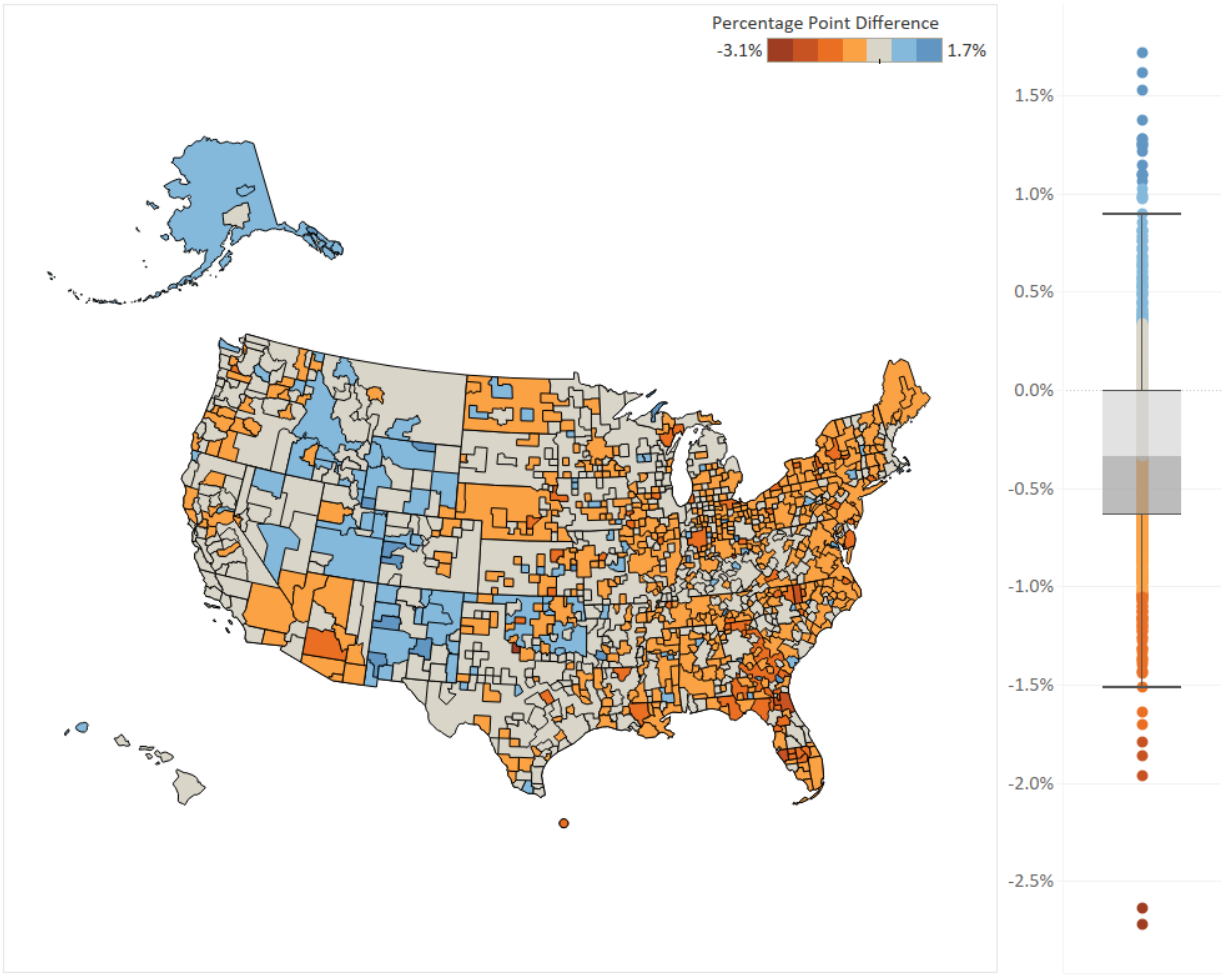
Source: ACS 2017-2021 5-year Estimates

Figure 2: Nonresponse Analysis of Household Administrative Data Income of Less than $40,000



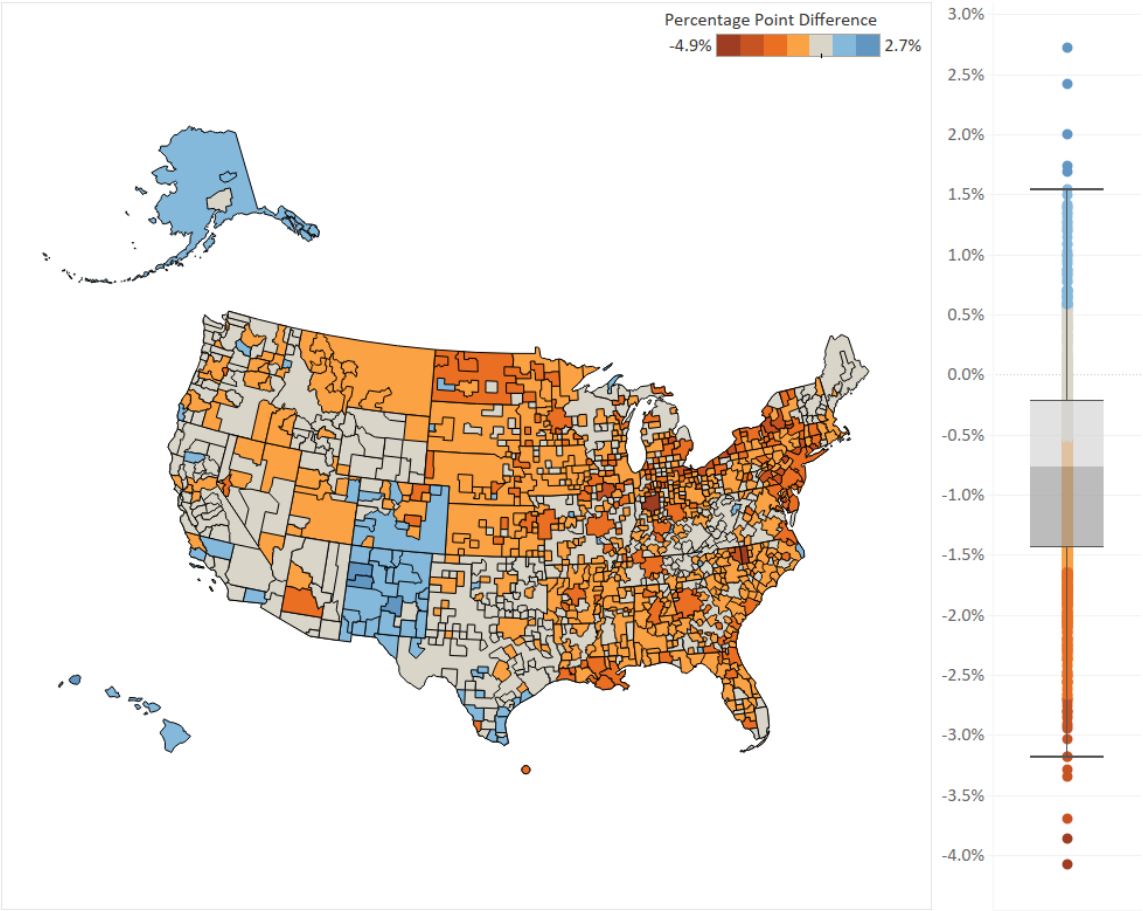
Source: ACS occupied housing units sampled in 2021 matched to administrative data. Figure presents bias estimate by CBSA group. This variable measures the percent of households with income between $1 and $40,000. For areas with small populations, additional disclosure protections were applied, which are described in Appendix A.2

Figure 3: Nonresponse Analysis of Presence of Household Member Aged Under 10 Years



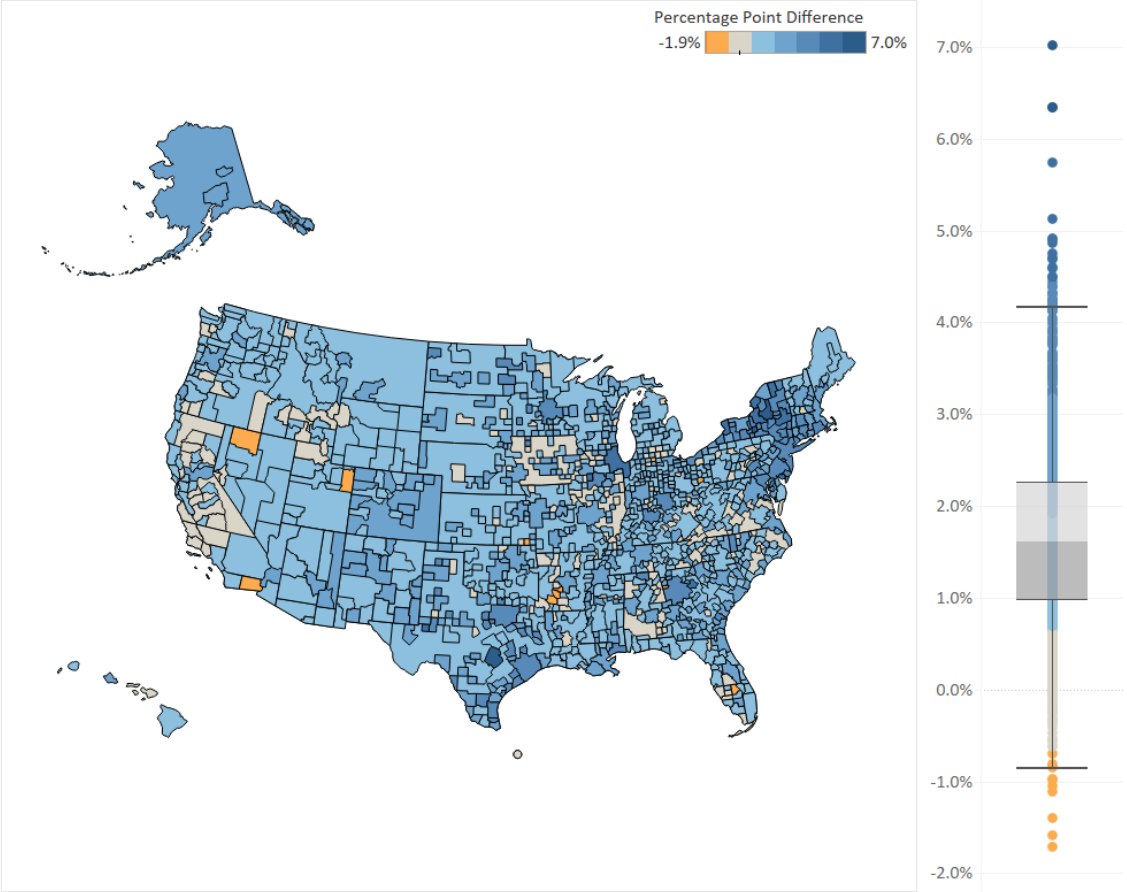
Source: ACS occupied housing units sampled in 2021 matched to administrative data. Figure presents bias estimate by CBSA group. For areas with small populations, additional disclosure protections were applied, which are described in Appendix A.2

Figure 4: Nonresponse Analysis of Presence of Household Member That is Not Non-Hispanic White Alone



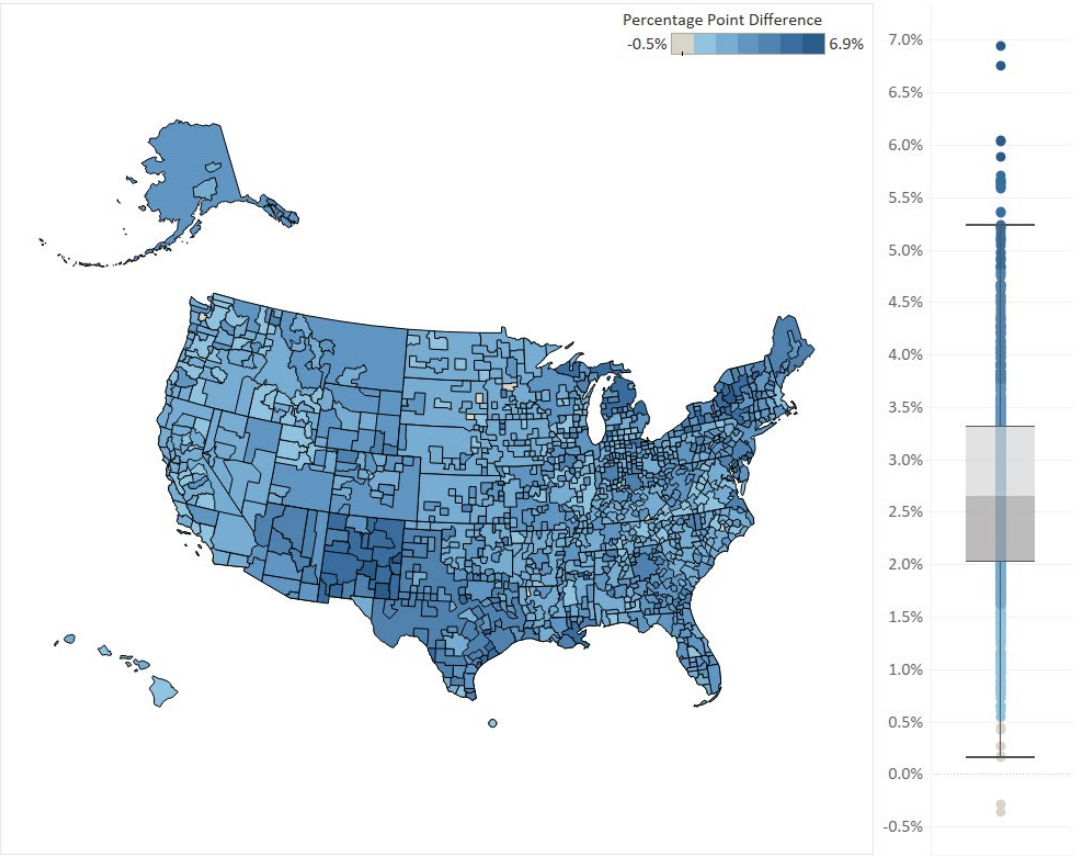
Source: ACS occupied housing units sampled in 2021 matched to administrative data. Figure presents bias estimate by CBSA group. For areas with small populations, additional disclosure protections were applied, which are described in Appendix A.2

Figure 5: Nonresponse Analysis of Presence of Single-Family Homes



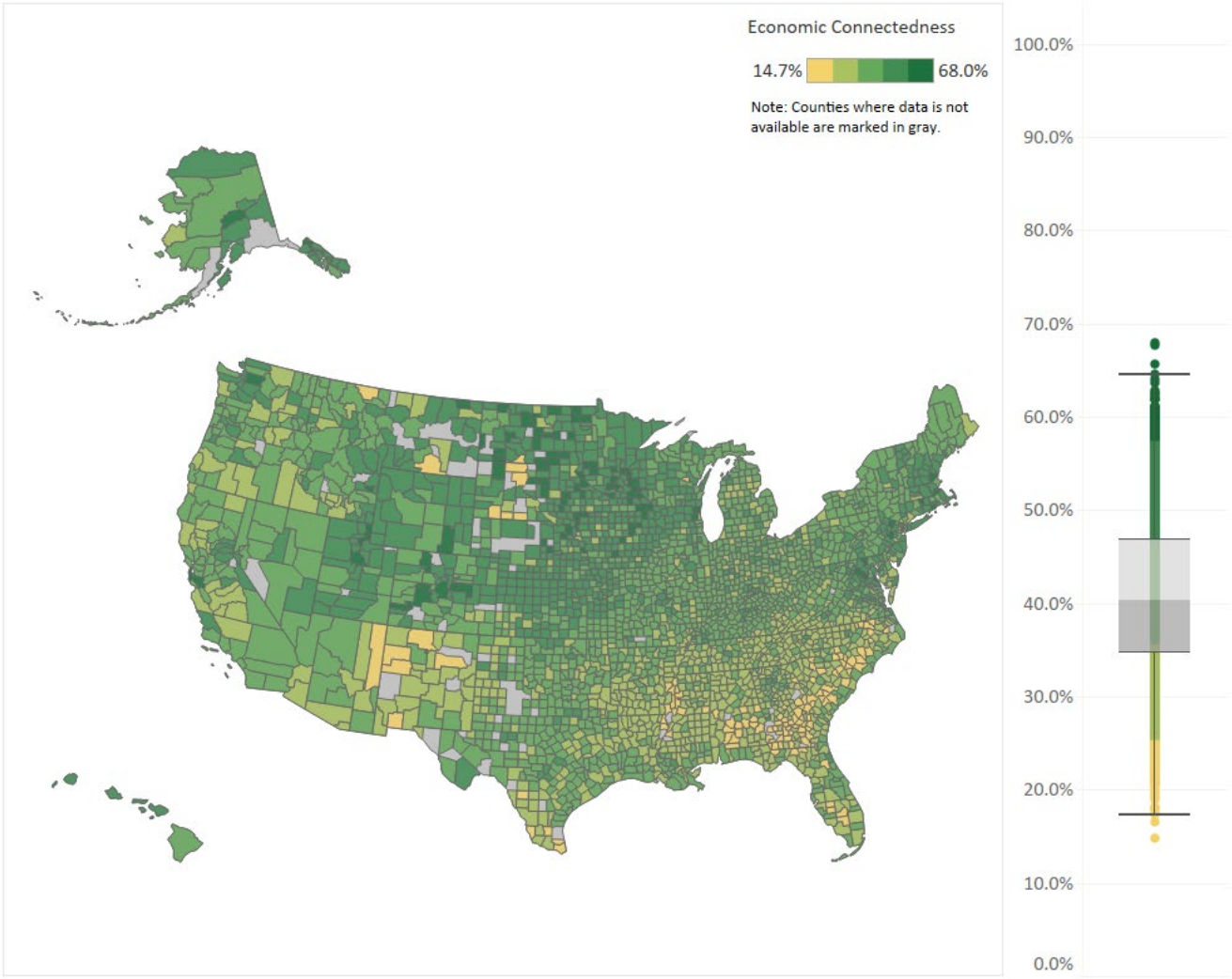
Source: ACS occupied housing units sampled in 2021 matched to administrative data. Figure presents bias estimate by CBSA group. For areas with small populations, additional disclosure protections were applied, which are described in Appendix A.2

Figure 6: Nonresponse Analysis of Presence of Household Member Aged Over 60 Years



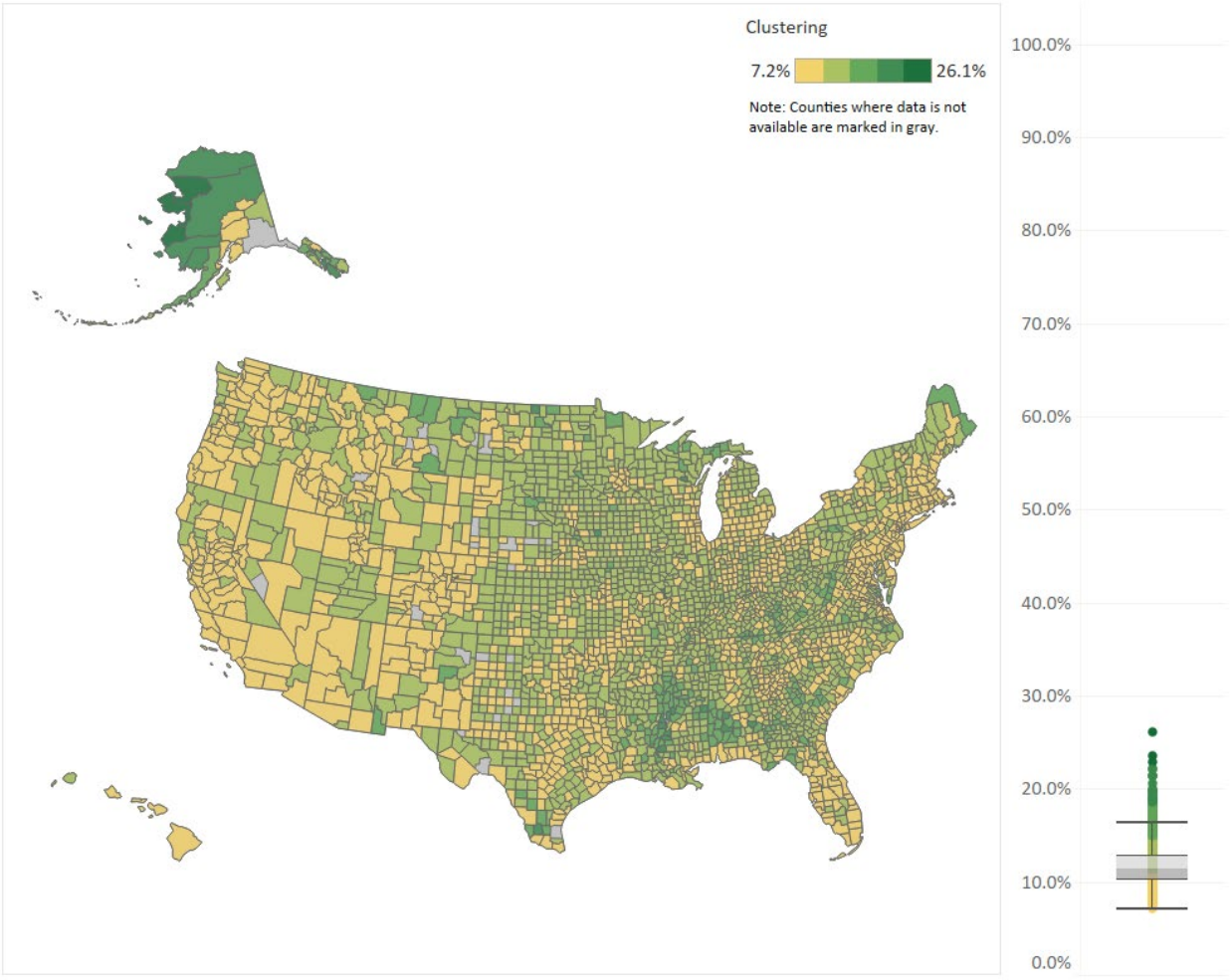
Source: ACS occupied housing units sampled in 2021 matched to administrative data. Figure presents bias estimate by CBSA group. For areas with small populations, additional disclosure protections were applied, which are described in Appendix A.2

Figure 7: Economic Connectedness



Source: Chetty et al. (2022). Data unavailable for some small counties.

Figure 8: Clustering



Source: Chetty et al. (2022). Data unavailable for some small counties.

Figures 1-8 show regional variation in these variables. ACS response rates are lower in the Southwest and higher in the Midwest. For the bias measures, Figure 2 presents regional variation in even the sign of the bias variable. In the Mid-Atlantic, respondents are less likely to be lower income compared to nonrespondents in the same area. In rural Southwest, on the other hand, respondents are more likely to be lower income. In Figure 3, in most areas of the country respondents are less likely to have young children, with a different sign for some areas

in the West. And for Figure 4 for whether there is a person of color in the household, Alaska, Hawaii, New Mexico, and parts of Colorado stand out with respondents being more likely to have such a person in the household, which is a different pattern from the rest of the county. One key takeaway is that these bias differences are variable specific. While some states like New Mexico stand out in many of the graphs, areas of New York seem to have higher bias only for the single-family home and household member over 60 variables. This reiterates a key point from Groves and Peytcheva (2008) who note that bias is variable specific, even within the same survey.

For the social capital variables, economic connectedness is lower in the South and higher in the rural Midwest. Clustering is higher with many counties in the South. While one can visually look at the Facebook variables to see the relationship with the ACS variables, we present correlations in Table 2 to provide a more formal summary. Table 2 presents the correlation between response rates and the Facebook variables for both CBSA and ACS estimation area county groups, as well as the correlation between the *absolute value* of the bias variables with response rates and the Facebook variables. We add response rates to these correlations to help put the magnitude of the Facebook correlations in context.

Table 2: Correlations of Social Capital with ACS Response Rates and Bias

	Response Rate		Economic Connectedness		Clustering	
CBSA	Correlation	P-Value	Correlation	P-Value	Correlation	P-Value
Response Rate			0.1789	<0.001	0.005711	0.8589
Income Over $130,000	-0.5582	<0.001	0.1148	<0.001	-0.122	<0.001
Income Under $40,000	-0.4879	<0.001	-0.1285	<0.001	0.149	<0.001
HH Member Over 60	-0.7942	<0.001	-0.0888	0.005621	0.04376	0.1731
HH Member Under 10	-0.3999	<0.001	-0.1633	<0.001	0.1822	<0.001
Any Non-Citizen in HH	-0.4313	<0.001	-0.09121	0.004447	-0.01158	0.7186

Any Persons of Color in HH	-0.5527	<0.001	-0.1555	<0.001	0.03482	0.2784
HH Member Disabled	-0.3743	<0.001	-0.2169	<0.001	0.2846	<0.001
Married Couple in HH	-0.7586	<0.001	-0.02815	0.3809	0.05135	0.1098
Structure Type is a Single-Family Home/Townhouse	-0.754	<0.001	0.02394	0.4562	-0.02166	0.5003
ACS Estimation Area						
Response Rate			0.2165	<0.001	0.03076	0.1564
Income Over $130,000	-0.5385	<0.001	0.09805	<0.001	-0.2275	<0.001
Income Under $40,000	-0.4147	<0.001	-0.0948	<0.001	0.1277	<0.001
HH Member Over 60	-0.7433	<0.001	-0.1295	<0.001	0.003943	0.8559
HH Member Under 10	-0.3765	<0.001	-0.1396	<0.001	0.1022	<0.001
Any Non-Citizen in HH	-0.3537	<0.001	-0.05708	0.008512	-0.09934	<0.001
Any Persons of Color in HH	-0.5	<0.001	-0.175	<0.001	0.003836	0.8598
HH Member Disabled	-0.3047	<0.001	-0.2243	<0.001	0.2665	<0.001
Married Couple in HH	-0.69	<0.001	-0.01107	0.6102	-0.1035	<0.001
Structure Type is a Single-Family Home/Townhouse	-0.5503	<0.001	-0.04253	0.04999	-0.08389	<0.001

Source: 2021 ACS, Administrative Data, and Facebook data from Chetty et al. (2022). Table presents Pearson correlation coefficients for these key variables. Variables other than responses rate are the absolute value of bias for a county group. Unit of observation is either a CBSA or ACS estimation area county group.

Overall, the table shows that areas with lower economic connectedness and higher clustering have higher bias. In other words, an area where people have more homogenous and insular social networks have higher bias. There are some caveats to note, though. First, the magnitude of most of the significant correlations are under 0.2 in absolute value, so the correlations are not high, especially relative to the correlations between nonresponse bias and response rates. The highest correlations are found with the disability variable, which are between 0.2 and 0.3 in absolute value. Interestingly, the correlation for disability with response rates is one of the lower values, with estimates of -0.3743 (CBSA) and -0.3047 (ACS estimation area). Additionally, the sign of the correlations between nonresponse bias and

economic connectedness are different for the income over $130,000 variable. In summary, these maps and correlations do show evidence of a relationship between area-level characteristics of people's social networks and area-level ACS nonresponse bias and response rates. In the next section, we perform some regression analysis to confirm this relationship is not being driven solely by other geographic characteristics.

4. Regressions

Next, we estimate the following regression model, where the key outcome is the absolute value of nonresponse bias for each variable v and county group c (CBSA or ACS estimation area). We specify the model as

$$\text{abs}\left(\bar{Y}_{v,c}^{R} - \bar{Y}_{v,c}\right) = \alpha_0 + \alpha_{1,v} RR_c + \alpha_{2,v} X_c + \epsilon_{c,v}$$

$$RR_c = \beta_0 + \beta_v X_c + u_{c,v}$$

where RR_c is the response rate in county group c and X_c is a vector of control variables. The goal with these two equations is to see how the social networking variables affect nonresponse bias through both A) their effect on response rates, as well as B) any residual effect after accounting for differences in response rates across county groups. To explain why there might be two separate effects, suppose an increase in the insularity of social networks causes people to be less likely to respond to surveys, but that this decrease is even more pronounced for certain demographic groups. In that case economic connectedness would be associate with higher bias, even after accounting for lower response rates in such areas. X_c includes all the Facebook variables described in the data section, as well as demographic and economic characteristics from the 2015-2019 ACS 5-year estimates. Given our bias measures are from

2021, picking this date range mitigates effects of our bias measures contributing to measurement error of our control variables. We also include a control of voting turnout rates, which are the 2020 presidential election vote counts divided by the 2015-2019 ACS 5-year estimates of the Citizen Voting Age Population.[5]

Table 3 presents the estimates of the unit response rate regression and Table 4 presents the coefficient estimates on all the dependent variables for the response rate, economic connectedness, and clustering coefficients. The full set of results are given in Appendix A.3. Additionally, Appendix A.4 presents estimation results of spatial regression models, which show largely similar results.

Table 3: Response Rate Regression

	CBSA		ACS Estimation Area	
	Estimate	S.E.	Estimate	S.E.
Intercept	69.76***	9.238	81.09***	5.577
Percent Children	0.045	0.151	-0.074	0.097
Percent Over 65	-0.083	0.119	0.065	0.080
Percent Hispanic	-0.272***	0.044	-0.176***	0.032
Percent Black	-0.088*	0.034	-0.062***	0.021
Percent Asian	-0.214*	0.099	-0.086	0.086
Percent Foreign Born	0.314**	0.106	0.023	0.075
Percent Poverty	0.180	0.104	-0.052	0.061
Percent College	-0.191***	0.058	-0.141***	0.035
Log Population Density	0.900*	0.357	0.545*	0.233
Economic Connectedness	0.136	0.076	0.175***	0.044
Clustering	0.057	0.288	0.061	0.201
Volunteering Rate	-0.159	0.097	-0.212**	0.066
Civic Organization	-36.86	75.540	-130.20***	36.850
Voter Turnout	0.198***	0.041	0.088***	0.030

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents regression results of

[5]Vote count totals come from MIT Election Data and Science Lab (2018).

response rates on explanatory variables for ACS households sampled in 2021. Hetroskedasticity-robust standard errors reported along with point estimates. Significance asterisks: *** p<.001 ** p<.01 * p<.05.

Table 4: Bias Regression Results for Select Explanatory Variables

	CBSA		ACS Estimation Area	
Dependent Variable	Estimate	S.E.	Estimate	S.E
Clustering				
Income Over $130,000	0.101***	0.0265	0.041*	0.0173
Income Under $40,000	0.128**	0.0413	0.052*	0.0246
HH Member Over 60	0.119*	0.0572	0.000	0.0366
HH Member Under 10	0.122*	0.0504	0.001	0.0205
Any Non-Citizen in HH	0.057**	0.0192	-0.001	0.0108
Any Persons of Color in HH	0.155**	0.0474	0.062*	0.0265
HH Member Disabled	0.103***	0.0218	0.048***	0.0140
Married Couple in HH	0.157**	0.0511	0.020	0.0342
Structure Type is a Single-Family Home/Townhouse	0.102*	0.0443	0.051	0.0265
Economic Connectedness				
Income Over $130,000	0.005	0.0082	0.002	0.0042
Income Under $40,000	0.001	0.0106	-0.002	0.0053
HH Member Over 60	0.026	0.0140	0.002	0.0079
HH Member Under 10	0.004	0.0072	0.001	0.0045
Any Non-Citizen in HH	0.001	0.0052	-0.006*	0.0028
Any Persons of Color in HH	-0.011	0.0131	-0.007	0.0063
HH Member Disabled	0.002	0.0054	0.003	0.0034
Married Couple in HH	0.006	0.0115	-0.009	0.0072
Structure Type is a Single-Family Home/Townhouse	0.024*	0.0116	0.007	0.0074
Response Rate				
Income Over $130,000	-0.073***	0.0054	-0.071***	0.0034
Income Under $40,000	-0.071***	0.0136	-0.059***	0.0070
HH Member Over 60	-0.249***	0.0159	-0.212***	0.0076
HH Member Under 10	-0.044***	0.0061	-0.048***	0.0039
Any Non-Citizen in HH	-0.029***	0.0054	-0.022***	0.0023
Any Persons of Color in HH	-0.094***	0.0156	-0.076***	0.0082
HH Member Disabled	-0.027***	0.0052	-0.025***	0.0024
Married Couple in HH	-0.210***	0.0130	-0.177***	0.0066
Structure Type is a Single-Family Home/Townhouse	-0.126***	0.0096	-0.114***	0.0060

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents regression results of explanatory variables on bias estimates from the 2021 ACS sample year. Hetroskedasticity-robust standard errors

reported along with point estimates. Results for all explanatory variables presented in Appendix A.4. Significance asterisks: *** p<.001 ** p<.01 * p<.05.

In Table 3 for the response rate regression, many of the race and ethnicity variables are significant. For the Hispanic coefficient in the right most set of columns, the estimate of -0.176 means that a 10 percentage point increase in the Hispanic population for an area is associated with a 1.76 percentage point lower response rate. For the social network variables, the clustering variable is insignificant in both regressions, and the economic connectedness variable is significant in the ACS estimation area regression. Of the people in the lower half of the income distribution, the economic connectedness variable represents the average share of their friends that are in the upper half of the income distribution. This coefficient estimate of 0.175 means that a 10 percentage point increase in the area-level share of low predicted SES individuals who are friends with high predicted SES individuals is associated with a 1.75 percentage point higher response rate for the area. Higher voter turnout rates in an area are also associated with higher ACS response rates.

In the ACS estimation area column, one odd result is the negative relationship between the volunteering and civic organization variables and response rates. Prior studies find that people who volunteer more are more likely to respond to surveys (e.g. Abraham et al. 2009). However, these proxies are more reflective of Facebook group membership than actual time spent on volunteering and civic engagement. Additionally, the results from Chetty et al. (2022) suggest lower explanatory power in these volunteering and civic organization variables, given they find weak correlations between these two measures and their primary outcome of interest, upward economic mobility. Thus, these odd results may potentially reflect issues with

the proxies rather than an actual negative relationship between volunteering and ACS response rates.

For the bias regressions, the economic connectedness coefficients are insignificant in the majority of the regressions. What stands out is the clustering variable, which is significant and positive in all of the CBSA regressions, and significant and positive in some of the ACS estimation area regressions. The estimate of 0.041 in the "Income over $130,000" row for the ACS estimation area regression means that if the area-level average percentage of a person's friends who are also friends with each other increases by 10 percentage points, this is associated with the bias of the high income variable being 0.4 percentage points higher. In other words, for areas where people have more insular networks, the bias is higher, even after controlling for response rate and other area characteristics.

Overall, the results in these regressions present similar results as Section 3. The significance of the results do vary by which county groups are used, so there are some issues with precision and model robustness in these regressions. But overall, the results suggests that the correlation coefficient estimates from Section 3 are not simply due to other area-level observable characteristics.

5. Conclusion

In this paper, we combine Facebook social networking data with ACS response rate and bias data to examine how social capital affects data quality. Our work makes use of a measure of social capital not available previously. Importantly, we can look at the effects on nonresponse bias, not just response rates. Overall, we find that areas where people have more insular and

sociodemographically homogeneous social networks have higher nonresponse bias in the ACS. The effect sizes are moderate and the significance of our estimates does vary by specification. But overall, we provide evidence that these social capital measures are related to survey participation and data quality.

As for implications, first, these results suggest possible unit nonresponse bias in any survey that measures characteristics about social networks. There is pre-existing evidence of nonresponse bias in other forms of behavior related to social capital (e.g. Abraham et al. 2009 with volunteering). Our results further suggest that survey-derived estimates about friendships and social networks, such as the number of close friends question on the General Social Survey, may be affected by nonresponse bias. Second, the results could suggest that having interviewers from the same community may result in higher cooperation rates. Various survey organizations already do this to reduce travel costs, but this is further evidence of the benefits of local interviewers. Third, these results suggest that finding a way to get a foothold into communities with insular social networks may be important for achieving a representative sample. Respondent-Driven Sampling (Heckathorn 1997) is a method for surveying hard-to-reach populations, so further study on how these ideas can be combined with traditional sampling methods may yield value for getting data from certain communities and people groups.

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Appendix

A.1 Administrative Data

In this appendix section, we provide details on how the administrative data are linked to the list of sampled ACS addresses. Our primary administrative data consists of tax data from the Internal Revenue Service (IRS). First, we use IRS Form 1040 individual income returns. The 1040 returns include income from the prior year as well as the identities of the first four children or family members claimed as dependents. Filing status from a 1040 return also gives us a proxy for marital status. Next, we use IRS information returns, which include W-2s from employers, 1099 forms such as the Form 1099-INT, and Form 1098.

To match survey and administrative records at the address level, we utilize the linking identifier in the Master Address File (MAF), which is the Census Bureau's frame file of all known living quarters and certain nonresidential addresses in the United States. The MAF is the sampling frame for the ACS, so all ACS observations have this linking identifier, called the MAFID. Data on whether a housing unit is a single-family home comes from the MAF. The IRS data are linked to the MAF using a probabilistic linking algorithm. Looking at 1040 data, Bee, Gathright, and Meyer (2015) find that about 90 percent of Form 1040 records matched to a MAFID. Because this match rate is less than 100 percent, there could be concerns that 1040s linked to the MAF may be different from unlinked 1040s. Bee, Gathright, and Meyer (2015) examine this as well and find that unmatched tax records tend to be in the extreme upper and lower ends of the income distribution. For our analyses, this finding implies that there are two possible reasons for why a household does not match to IRS data. It could be that this household did not file their taxes or receive any other tax form (among those provided to the Census Bureau) from an

employer or other third party. However, it could also be the case that they did have one of these tax forms, but had certain characteristics resulting in their tax form failing to match to the MAF.

Construction of Household Roster

After linking several administrative datasets to the ACS by address (MAFID), our second step is to use the administrative data to construct a household roster for each address that we then use as a basis for matching additional person-level datasets. Our primary sources for constructing a household roster are the IRS 1040 returns and the IRS information returns (e.g., Forms 1098, 1099, and W-2). Importantly, the dependent information on the 1040s gives us information about children in the household. Using the information returns in addition to the 1040s is useful for picking up individuals who do not have any filing requirements. For example, retired individuals who only receive Old-Age, Survivors, and Disability Insurance (OASDI) benefits typically do not have any taxable income. These individuals then might not file an IRS 1040 tax return, but they would receive an SSA-1099, so using the information returns helps us add them to household rosters.

However, some individuals in the United States do not appear in any IRS data available to the Census Bureau. To locate some of these individuals, we utilize two additional sources when constructing the household rosters. First, we use location information from Census Bureau's Master Address File Auxiliary Reference File (MAFARF). The MAFARF contains residency information from IRS data and other administrative data sources, such as data sets from the United States Department of Housing and Urban Development (HUD). Thus, the MAFARF picks

up some additional people who do not have any taxable income. Second, we use data on Supplemental Security Income (SSI) recipients from SSA's Supplemental Security Record (SSR). Since SSI is not taxable, individuals who only receive SSI income would not appear in IRS data. In addition, the SSR is not used in construction of the MAFARF. Thus, the MAFARF and SSR allow us to pick up additional people who do not appear in IRS data.

Given the household roster, we link additional datasets to infer more information about these households. We use demographic data from the SSA's Numident file, the 2010 Decennial Census, and the 2020 Decennial Census. Both the Numident and decennial data contain information on an individual's age and race. If race data are available from both sources for a person, we use decennial race, and give primacy to 2020 data over 2010 data. The decennial census gives us information on Hispanic origin, while the Numident data contain information on citizenship and foreign-born status.

Both these datasets are linked to the IRS-based household rosters at the individual level using the Census Bureau's Person Identification Validation System (PVS), as described by Wagner and Layne (2014). This procedure matches both survey data and administrative data to a master reference file. Individuals who are matched are given an identifier called a Protected Identification Key (PIK), which acts as an anonymized Social Security number that can be used to link administrative datasets and surveys. For the 2010 Census, about 90 percent of individuals were assigned a PIK (Wagner and Layne, 2014).[6] Bond et al. (2014) argue that

[6] The 2010 and 2020 Census is also linked to the MAF. Because many people have changed their residential location since a decennial census, however, we link decennial data at the person level instead, to make sure we are capturing the characteristics of the current household.

inability to assign a reliable PIK is nonrandom. They use 2009 ACS data to document that young children, minorities, immigrants, recent movers, low-income individuals, and non-employed individuals are less likely to receive a PIK.

Once the Numident and decennial census data are matched at the individual level, the data are then aggregated to the household level to create address-level measures for comparing respondent and nonrespondent households. For example, our measure of the presence of a household member over age 60 comes from taking the list of people given on the tax forms matched to this address, and then matching these people to the Numident file to get the year of birth for all household members. Additionally, we match OASDI program benefit data for SSA's Master Beneficiary Record (MBR) and the Payment History Update System (PHUS) at the person level using PIK, which gives us disability information.

A.2 Disclosure Protections on Maps

For the nonresponse bias maps presented in Section 3, some of the CBSAs have small populations. Current disclosure policy at the U.S. Census Bureau requires special disclosure procedures when releasing estimates of areas that have a populations smaller than the smallest congressional district. In these maps, the plotted estimate for such areas is a weighted average of the actual estimate and the overall state average. The weights are defined as the ratio of the standard error of the estimate for Wyoming divided by the standard error for the underlying estimate, with the standard error typically increasing the smaller the population is. We enforced a maximum value of the weight to ensure that the state average always contributes some amount to the plotted estimate. Because of these disclosure protections, the

appearance that areas within a state have similar values is somewhat artificial due to these protections.

A.3 Full Bias Results

Table 5: Bias Regression Results for Additional Variables

		CBSA		ACS Estimation Area	
Dependent Variable	Coefficient	Estimate	S.E.	Estimate	S.E
Income Over $130,000	Intercept	5.362***	0.9170	6.493***	0.5419
Income Under $40,000	Intercept	5.150***	1.4360	5.814***	0.8423
HH Member Over 60	Intercept	20.100***	1.9070	20.710***	1.1210
HH Member Under 10	Intercept	3.367*	1.4420	5.046***	0.7628
Any Non-Citizen in HH	Intercept	3.653**	1.3580	3.174***	0.4886
Any Persons of Color in HH	Intercept	8.918***	2.3830	6.926***	1.2210
HH Member Disabled	Intercept	2.748*	1.2790	2.521***	0.5145
Married Couple in HH	Intercept	19.310***	1.5820	19.250***	1.0850
Structure Type is a Single-Family Home/Townhouse	Intercept	9.869***	1.5880	11.470***	0.9817
Income Over $130,000	Civic Organization	0.773	7.5530	6.888	3.8270
Income Under $40,000	Civic Organization	0.694	10.4000	8.367	4.9150
HH Member Over 60	Civic Organization	12.080	15.3500	17.290*	7.0360
HH Member Under 10	Civic Organization	-3.798	9.2260	0.867	4.4770
Any Non-Citizen in HH	Civic Organization	-3.455	8.0620	0.528	2.7140
Any Persons of Color in HH	Civic Organization	-28.440*	12.8800	-9.138	5.4770
HH Member Disabled	Civic Organization	-9.686	7.4780	-2.915	2.7310
Married Couple in HH	Civic Organization	-13.570	12.9500	6.938	6.9880
Structure Type is a Single-Family Home/Townhouse	Civic Organization	-0.088	12.8100	-5.023	6.3420
Income Over $130,000	Log Population Density	0.089*	0.0350	0.070**	0.0219
Income Under $40,000	Log Population Density	-0.060	0.0518	-0.063*	0.0278
HH Member Over 60	Log Population Density	0.175**	0.0670	0.137**	0.0418
HH Member Under 10	Log Population Density	-0.015	0.0360	-0.050	0.0269
Any Non-Citizen in HH	Log Population Density	-0.092**	0.0296	-0.080***	0.0187
Any Persons of Color in HH	Log Population Density	0.015	0.0599	0.002	0.0363
HH Member Disabled	Log Population Density	-0.055	0.0295	-0.014	0.0153

Married Couple in HH	Log Population Density	0.091	0.0557	0.062	0.0405
Structure Type is a Single-Family Home/Townhouse	Log Population Density	0.183***	0.0540	0.096**	0.0359
Income Over $130,000	Percent Asian	-0.010	0.0128	-0.008	0.0113
Income Under $40,000	Percent Asian	-0.004	0.0117	0.011	0.0099
HH Member Over 60	Percent Asian	-0.056**	0.0187	-0.003	0.0149
HH Member Under 10	Percent Asian	-0.018	0.0096	-0.012	0.0116
Any Non-Citizen in HH	Percent Asian	-0.032*	0.0127	-0.020*	0.0101
Any Persons of Color in HH	Percent Asian	0.013	0.0236	0.013	0.0180
HH Member Disabled	Percent Asian	-0.013*	0.0063	-0.008	0.0051
Married Couple in HH	Percent Asian	-0.006	0.0229	0.025	0.0175
Structure Type is a Single-Family Home/Townhouse	Percent Asian	-0.008	0.0181	0.001	0.0146
Income Over $130,000	Percent Black	0.001	0.0035	0.003	0.0019
Income Under $40,000	Percent Black	-0.004	0.0050	0.000	0.0028
HH Member Over 60	Percent Black	0.000	0.0071	-0.001	0.0042
HH Member Under 10	Percent Black	-0.004	0.0050	0.003	0.0025
Any Non-Citizen in HH	Percent Black	-0.001	0.0021	-0.001	0.0011
Any Persons of Color in HH	Percent Black	-0.003	0.0062	0.003	0.0029
HH Member Disabled	Percent Black	0.003	0.0031	0.003	0.0018
Married Couple in HH	Percent Black	-0.017**	0.0059	-0.009*	0.0036
Structure Type is a Single-Family Home/Townhouse	Percent Black	-0.003	0.0057	-0.003	0.0033
Income Over $130,000	Percent Children	0.024	0.0149	0.008	0.0090
Income Under $40,000	Percent Children	0.042*	0.0206	0.018	0.0115
HH Member Over 60	Percent Children	0.033	0.0303	-0.019	0.0196
HH Member Under 10	Percent Children	0.029	0.0177	0.026*	0.0122

Any Non-Citizen in HH	Percent Children	0.005	0.0231	0.013*	0.0068
Any Persons of Color in HH	Percent Children	-0.007	0.0319	0.027	0.0142
HH Member Disabled	Percent Children	-0.011	0.0212	-0.004	0.0080
Married Couple in HH	Percent Children	0.015	0.0238	-0.027	0.0174
Structure Type is a Single-Family Home/Townhouse	Percent Children	0.017	0.0266	-0.016	0.0151
Income Over $130,000	Percent College	0.027***	0.0064	0.018***	0.0036
Income Under $40,000	Percent College	0.007	0.0081	0.000	0.0041
HH Member Over 60	Percent College	-0.009	0.0113	-0.011	0.0064
HH Member Under 10	Percent College	0.001	0.0071	-0.010*	0.0041
Any Non-Citizen in HH	Percent College	0.001	0.0047	0.003	0.0024
Any Persons of Color in HH	Percent College	0.011	0.0100	0.009	0.0050
HH Member Disabled	Percent College	-0.003	0.0047	-0.010***	0.0026
Married Couple in HH	Percent College	0.028**	0.0104	0.009	0.0062
Structure Type is a Single-Family Home/Townhouse	Percent College	0.012	0.0097	0.017**	0.0059
Income Over $130,000	Percent Foreign Born	0.028*	0.0111	0.026*	0.0113
Income Under $40,000	Percent Foreign Born	0.003	0.0138	-0.013	0.0095
HH Member Over 60	Percent Foreign Born	0.020	0.0230	-0.053**	0.0161
HH Member Under 10	Percent Foreign Born	0.004	0.0115	0.003	0.0107
Any Non-Citizen in HH	Percent Foreign Born	0.053***	0.0137	0.037***	0.0082
Any Persons of Color in HH	Percent Foreign Born	-0.027	0.0231	-0.031	0.0161
HH Member Disabled	Percent Foreign Born	0.013*	0.0066	0.002	0.0046
Married Couple in HH	Percent Foreign Born	0.000	0.0214	-0.035*	0.0158
Structure Type is a Single-Family Home/Townhouse	Percent Foreign Born	0.018	0.0209	-0.008	0.0153
Income Over $130,000	Percent Hispanic	-0.005	0.0040	-0.007*	0.0035
Income Under $40,000	Percent Hispanic	0.001	0.0058	0.003	0.0044
HH Member Over 60	Percent Hispanic	-0.008	0.0094	0.010	0.0071
HH Member Under 10	Percent Hispanic	0.003	0.0049	-0.004	0.0039
Any Non-Citizen in HH	Percent Hispanic	0.000	0.0040	-0.002	0.0025
Any Persons of Color in HH	Percent Hispanic	0.020*	0.0093	0.019**	0.0070
HH Member Disabled	Percent Hispanic	0.001	0.0029	-0.001	0.0020
Married Couple in HH	Percent Hispanic	-0.014	0.0083	0.000	0.0062
Structure Type is a Single-Family Home/Townhouse	Percent Hispanic	-0.014	0.0076	-0.006	0.0058

Income Over $130,000	Percent Over 65	-0.009	0.0108	-0.016*	0.0074
Income Under $40,000	Percent Over 65	0.014	0.0167	-0.003	0.0099
HH Member Over 60	Percent Over 65	0.067**	0.0245	0.038*	0.0151
HH Member Under 10	Percent Over 65	0.019	0.0123	0.000	0.0100
Any Non-Citizen in HH	Percent Over 65	-0.007	0.0102	-0.006	0.0062
Any Persons of Color in HH	Percent Over 65	-0.023	0.0192	-0.001	0.0106
HH Member Disabled	Percent Over 65	0.007	0.0108	0.006	0.0061
Married Couple in HH	Percent Over 65	-0.013	0.0171	-0.034*	0.0137
Structure Type is a Single-Family Home/Townhouse	Percent Over 65	-0.030	0.0178	-0.029*	0.0127
Income Over $130,000	Percent Poverty	-0.033***	0.0096	-0.035***	0.0053
Income Under $40,000	Percent Poverty	-0.002	0.0139	-0.008	0.0072
HH Member Over 60	Percent Poverty	-0.025	0.0200	-0.027*	0.0130
HH Member Under 10	Percent Poverty	-0.009	0.0097	-0.004	0.0068
Any Non-Citizen in HH	Percent Poverty	-0.026*	0.0106	-0.021***	0.0042
Any Persons of Color in HH	Percent Poverty	-0.027	0.0147	-0.013	0.0075
HH Member Disabled	Percent Poverty	-0.004	0.0095	0.005	0.0048
Married Couple in HH	Percent Poverty	-0.043*	0.0170	-0.050***	0.0111
Structure Type is a Single-Family Home/Townhouse	Percent Poverty	0.010	0.0162	-0.005	0.0099
Income Over $130,000	Volunteering Rate	0.005	0.0119	-0.005	0.0066
Income Under $40,000	Volunteering Rate	-0.008	0.0155	-0.003	0.0093
HH Member Over 60	Volunteering Rate	0.001	0.0250	-0.001	0.0129
HH Member Under 10	Volunteering Rate	-0.019	0.0125	-0.004	0.0077
Any Non-Citizen in HH	Volunteering Rate	-0.006	0.0098	-0.009*	0.0045
Any Persons of Color in HH	Volunteering Rate	-0.016	0.0167	-0.021*	0.0093
HH Member Disabled	Volunteering Rate	0.008	0.0107	0.001	0.0059
Married Couple in HH	Volunteering Rate	-0.015	0.0203	0.000	0.0122
Structure Type is a Single-Family Home/Townhouse	Volunteering Rate	-0.017	0.0214	-0.018	0.0125
Income Over $130,000	Voter Turnout	-0.005	0.0046	-0.001	0.0029
Income Under $40,000	Voter Turnout	-0.006	0.0061	-0.002	0.0038
HH Member Over 60	Voter Turnout	-0.004	0.0085	-0.001	0.0057
HH Member Under 10	Voter Turnout	-0.011*	0.0052	0.000	0.0045
Any Non-Citizen in HH	Voter Turnout	-0.011**	0.0041	-0.004*	0.0022
Any Persons of Color in HH	Voter Turnout	0.006	0.0073	0.001	0.0040

HH Member Disabled	Voter Turnout	-0.008*	0.0035	-0.002	0.0022
Married Couple in HH	Voter Turnout	-0.007	0.0078	0.003	0.0054
Structure Type is a Single-Family Home/Townhouse	Voter Turnout	-0.003	0.0080	-0.004	0.0051

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents regression results of explanatory variables on bias estimates from the 2021 ACS sample year. Hetroskedasticity-robust standard errors reported along with point estimates. Significance asterisks: *** p<.001 ** p<.01 * p<.05.

A.4 Spatial Regression

The graphs from Section 3 show neighboring areas having similar response rates, bias values, and social capital values. This could be suggestive that the observations are not independent of each other, violating the independence assumption in the prior regression model. Because of this, we also estimate a spatial regression model as a robustness check. We estimate both a spatial lag and spatial error model. We use queen contiguity to construct neighbors for our geographies, assigning equal weights to all neighbors. Because our neighbors are constructed with shared borders, we exclude all of Alaska and Hawaii, and a few counties in Washington State and Massachusetts that are islands.

The following tables show estimates from these models. Because the coefficients from the spatial lag model do not give marginal effects (that is, the estimate of how much the dependent variables changes by a one unit change in the explanatory variable), we present the marginal effects instead of the coefficient estimates. Note that there is no marginal effect for the intercept, which is why those rows are flagged as NA in the tables below. Overall, the results are similar. Some differences are for the response rate regressions, where the economic clustering variable is now significant for both the CBSA and the ACS estimation area county groupings.

Table 6: Response Rate Regression with Spatial Error Model

	CBSA		ACS Estimation Area	
	Estimate	S.E.	Estimate	S.E.
Intercept	67.50***	6.785	81.15***	4.505
Percent Children	0.045	0.136	-0.135	0.082
Percent Over 65	-0.180	0.100	-0.069	0.067
Percent Hispanic	-0.209***	0.039	-0.056	0.034
Percent Black	-0.098**	0.035	-0.093***	0.021
Percent Asian	-0.287	0.154	-0.002	0.103
Percent Foreign Born	0.196*	0.099	-0.054	0.076
Percent Poverty	0.098	0.072	-0.141**	0.047
Percent College	-0.172**	0.054	-0.078**	0.030
Log Population Density	1.040**	0.364	0.539*	0.231
Economic Connectedness	0.173**	0.066	0.155***	0.040
Clustering	0.345	0.245	0.207	0.151
Volunteering Rate	-0.176	0.095	-0.122	0.065
Civic Organization	38.83	59.770	-66.56*	30.010
Voter Turnout	0.184***	0.048	0.098**	0.030

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents spatial error regression model results of explanatory variables on response rates for ACS households sampled in 2021. Significance asterisks: *** p<.001 ** p<.01 * p<.05.

Table 7: Response Rate Regression with Spatial Lag Model

	CBSA		ACS Estimation Area	
	Estimate	S.E.	Estimate	S.E.
Intercept	NA	NA	NA	NA
Percent Children	0.179	0.123	-0.254***	0.076
Percent Over 65	-0.110	0.095	0.043	0.064
Percent Hispanic	-0.359***	0.029	-0.101***	0.019
Percent Black	-0.143***	0.027	-0.107***	0.016
Percent Asian	-0.501***	0.141	0.173	0.096
Percent Foreign Born	0.371***	0.079	-0.231***	0.054
Percent Poverty	0.350***	0.067	-0.239***	0.045
Percent College	-0.234***	0.050	-0.120***	0.031
Log Population Density	1.121***	0.268	-0.109	0.171
Economic Connectedness	0.251***	0.057	0.228***	0.034
Clustering	-0.066	0.220	0.226	0.129
Volunteering Rate	-0.314***	0.091	-0.288***	0.059
Civic Organization	-18.82	60.660	-335.50***	32.460

Voter Turnout	0.366***	0.038	0.190***	0.025

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents spatial lag regression model results of explanatory variables on response rates for ACS households sampled in 2021. Displayed coefficients are the marginal effects. Significance asterisks: *** p<.001 ** p<.01 * p<.05.

Table 8: Bias Regression Results for Spatial Error Model

		CBSA		ACS Estimation Area	
Dependent Variable	Coefficient	Estimate	S.E.	Estimate	S.E
Income Over $130,000	Intercept	5.414***	0.7498	6.460***	0.5196
Income Under $40,000	Intercept	4.831***	1.0140	6.025***	0.6708
HH Member Over 60	Intercept	20.180***	1.4550	21.070***	0.9656
HH Member Under 10	Intercept	3.350***	0.8876	4.889***	0.6268
Any Non-Citizen in HH	Intercept	3.814***	0.5986	2.860***	0.3519
Any Persons of Color in HH	Intercept	9.523***	1.2680	6.913***	0.7647
HH Member Disabled	Intercept	2.820***	0.5901	2.745***	0.4114
Married Couple in HH	Intercept	19.210***	1.3120	19.020***	0.8958
Structure Type is a Single-Family Home/Townhouse	Intercept	9.725***	1.3390	11.700***	0.9127
Income Over $130,000	Civic Organization	-0.819	6.7420	7.839*	3.6630
Income Under $40,000	Civic Organization	3.486	9.6310	9.228	4.9030
HH Member Over 60	Civic Organization	8.503	13.0100	15.930*	6.8480
HH Member Under 10	Civic Organization	-4.091	8.2220	1.087	4.5190
Any Non-Citizen in HH	Civic Organization	-3.237	5.5180	0.234	2.5430
Any Persons of Color in HH	Civic Organization	-29.310*	11.6200	-8.825	5.5070
HH Member Disabled	Civic Organization	-8.836	5.5070	-3.162	2.9230
Married Couple in HH	Civic Organization	-10.800	12.1000	7.436	6.4840
Structure Type is a Single-Family Home/Townhouse	Civic Organization	1.159	12.3200	-4.718	6.4290
Income Over $130,000	Clustering	0.114***	0.0246	0.047**	0.0154
Income Under $40,000	Clustering	0.137***	0.0325	0.069***	0.0196
HH Member Over 60	Clustering	0.098*	0.0478	0.010	0.0286
HH Member Under 10	Clustering	0.122***	0.0287	0.001	0.0184
Any Non-Citizen in HH	Clustering	0.059**	0.0194	0.004	0.0103
Any Persons of Color in HH	Clustering	0.170***	0.0412	0.045*	0.0225
HH Member Disabled	Clustering	0.107***	0.0190	0.058***	0.0122
Married Couple in HH	Clustering	0.174***	0.0425	0.046	0.0263
Structure Type is a Single-Family Home/Townhouse	Clustering	0.106*	0.0434	0.057*	0.0271
Income Over $130,000	Economic Connectedness	-0.002	0.0065	0.001	0.0040
Income Under $40,000	Economic Connectedness	0.001	0.0084	-0.003	0.0050

HH Member Over 60	Economic Connectedness	0.031*	0.0127	0.001	0.0074
HH Member Under 10	Economic Connectedness	0.006	0.0075	0.002	0.0047
Any Non-Citizen in HH	Economic Connectedness	0.000	0.0051	-0.006*	0.0027
Any Persons of Color in HH	Economic Connectedness	-0.016	0.0109	-0.007	0.0058
HH Member Disabled	Economic Connectedness	0.002	0.0050	0.002	0.0031
Married Couple in HH	Economic Connectedness	0.005	0.0112	-0.010	0.0067
Structure Type is a Single-Family Home/Townhouse	Economic Connectedness	0.023*	0.0114	0.003	0.0070
Income Over $130,000	Log Population Density	0.071*	0.0326	0.063**	0.0212
Income Under $40,000	Log Population Density	-0.059	0.0393	-0.080**	0.0263
HH Member Over 60	Log Population Density	0.119	0.0640	0.108**	0.0392
HH Member Under 10	Log Population Density	-0.023	0.0362	-0.047	0.0250
Any Non-Citizen in HH	Log Population Density	-0.092***	0.0247	-0.070***	0.0140
Any Persons of Color in HH	Log Population Density	-0.012	0.0529	0.007	0.0305
HH Member Disabled	Log Population Density	-0.068**	0.0237	-0.028	0.0167
Married Couple in HH	Log Population Density	0.073	0.0539	0.058	0.0355
Structure Type is a Single-Family Home/Townhouse	Log Population Density	0.176**	0.0554	0.083*	0.0373
Income Over $130,000	Percent Asian	0.001	0.0162	-0.006	0.0110
Income Under $40,000	Percent Asian	0.029	0.0215	0.031*	0.0141
HH Member Over 60	Percent Asian	-0.042	0.0314	0.008	0.0203
HH Member Under 10	Percent Asian	-0.013	0.0190	-0.022	0.0132
Any Non-Citizen in HH	Percent Asian	-0.052***	0.0129	-0.038***	0.0074
Any Persons of Color in HH	Percent Asian	-0.001	0.0273	-0.010	0.0161
HH Member Disabled	Percent Asian	0.000	0.0126	-0.002	0.0087
Married Couple in HH	Percent Asian	0.032	0.0282	0.037	0.0188
Structure Type is a Single-Family Home/Townhouse	Percent Asian	0.020	0.0288	0.004	0.0193
Income Over $130,000	Percent Black	-0.002	0.0033	0.002	0.0019
Income Under $40,000	Percent Black	-0.003	0.0043	-0.001	0.0024

HH Member Over 60	Percent Black	0.002	0.0065	-0.002	0.0035
HH Member Under 10	Percent Black	-0.003	0.0038	0.003	0.0023
Any Non-Citizen in HH	Percent Black	-0.002	0.0026	-0.001	0.0013
Any Persons of Color in HH	Percent Black	-0.004	0.0055	0.004	0.0028
HH Member Disabled	Percent Black	0.003	0.0025	0.003	0.0015
Married Couple in HH	Percent Black	-0.018**	0.0057	-0.011***	0.0032
Structure Type is a Single-Family Home/Townhouse	Percent Black	-0.004	0.0058	-0.004	0.0034
Income Over $130,000	Percent Children	0.022	0.0140	0.009	0.0089
Income Under $40,000	Percent Children	0.048*	0.0189	0.019	0.0115
HH Member Over 60	Percent Children	0.040	0.0272	-0.015	0.0165
HH Member Under 10	Percent Children	0.031	0.0165	0.030**	0.0107
Any Non-Citizen in HH	Percent Children	0.002	0.0112	0.018**	0.0060
Any Persons of Color in HH	Percent Children	-0.011	0.0237	0.026*	0.0131
HH Member Disabled	Percent Children	-0.010	0.0110	-0.004	0.0070
Married Couple in HH	Percent Children	0.021	0.0245	-0.019	0.0154
Structure Type is a Single-Family Home/Townhouse	Percent Children	0.022	0.0250	-0.015	0.0156
Income Over $130,000	Percent College	0.031***	0.0057	0.018***	0.0034
Income Under $40,000	Percent College	0.004	0.0080	-0.001	0.0044
HH Member Over 60	Percent College	-0.007	0.0111	-0.009	0.0063
HH Member Under 10	Percent College	-0.001	0.0069	-0.009*	0.0041
Any Non-Citizen in HH	Percent College	0.002	0.0046	0.005*	0.0023
Any Persons of Color in HH	Percent College	0.016	0.0098	0.012*	0.0050
HH Member Disabled	Percent College	-0.005	0.0046	-0.009***	0.0027
Married Couple in HH	Percent College	0.029**	0.0102	0.011	0.0059
Structure Type is a Single-Family Home/Townhouse	Percent College	0.010	0.0104	0.018**	0.0059
Income Over $130,000	Percent Foreign Born	0.022*	0.0098	0.027***	0.0071
Income Under $40,000	Percent Foreign Born	-0.002	0.0126	-0.016	0.0088
HH Member Over 60	Percent Foreign Born	0.014	0.0192	-0.051***	0.0131
HH Member Under 10	Percent Foreign Born	0.003	0.0113	0.004	0.0084
Any Non-Citizen in HH	Percent Foreign Born	0.060***	0.0077	0.041***	0.0047
Any Persons of Color in HH	Percent Foreign Born	-0.021	0.0163	-0.026*	0.0102
HH Member Disabled	Percent Foreign Born	0.012	0.0075	0.003	0.0056
Married Couple in HH	Percent Foreign Born	-0.007	0.0168	-0.036**	0.0119

Structure Type is a Single-Family Home/Townhouse	Percent Foreign Born	0.012	0.0172	-0.003	0.0125
Income Over $130,000	Percent Hispanic	-0.005	0.0035	-0.008**	0.0027
Income Under $40,000	Percent Hispanic	0.003	0.0043	0.002	0.0032
HH Member Over 60	Percent Hispanic	-0.008	0.0069	0.009	0.0049
HH Member Under 10	Percent Hispanic	0.003	0.0040	-0.004	0.0031
Any Non-Citizen in HH	Percent Hispanic	-0.002	0.0027	-0.002	0.0017
Any Persons of Color in HH	Percent Hispanic	0.018**	0.0058	0.018***	0.0038
HH Member Disabled	Percent Hispanic	0.001	0.0026	-0.002	0.0021
Married Couple in HH	Percent Hispanic	-0.013*	0.0059	0.000	0.0044
Structure Type is a Single-Family Home/Townhouse	Percent Hispanic	-0.013*	0.0060	-0.007	0.0047
Income Over $130,000	Percent Over 65	-0.014	0.0107	-0.018*	0.0075
Income Under $40,000	Percent Over 65	0.016	0.0147	-0.009	0.0097
HH Member Over 60	Percent Over 65	0.077***	0.0207	0.035*	0.0139
HH Member Under 10	Percent Over 65	0.020	0.0128	0.001	0.0091
Any Non-Citizen in HH	Percent Over 65	-0.011	0.0086	-0.005	0.0051
Any Persons of Color in HH	Percent Over 65	-0.031	0.0182	0.004	0.0111
HH Member Disabled	Percent Over 65	0.006	0.0085	0.002	0.0059
Married Couple in HH	Percent Over 65	-0.011	0.0189	-0.037**	0.0130
Structure Type is a Single-Family Home/Townhouse	Percent Over 65	-0.030	0.0193	-0.031*	0.0131
Income Over $130,000	Percent Poverty	-0.031***	0.0077	-0.033***	0.0053
Income Under $40,000	Percent Poverty	-0.004	0.0105	-0.009	0.0068
HH Member Over 60	Percent Poverty	-0.021	0.0149	-0.027**	0.0098
HH Member Under 10	Percent Poverty	-0.008	0.0092	-0.003	0.0064
Any Non-Citizen in HH	Percent Poverty	-0.026***	0.0062	-0.018***	0.0036
Any Persons of Color in HH	Percent Poverty	-0.030*	0.0131	-0.012	0.0078
HH Member Disabled	Percent Poverty	-0.005	0.0061	0.004	0.0042
Married Couple in HH	Percent Poverty	-0.044**	0.0136	-0.048***	0.0091
Structure Type is a Single-Family Home/Townhouse	Percent Poverty	0.012	0.0138	-0.004	0.0093
Income Over $130,000	Response Rate	-0.071***	0.0034	-0.072***	0.0023
Income Under $40,000	Response Rate	-0.072***	0.0045	-0.062***	0.0030
HH Member Over 60	Response Rate	-0.248***	0.0066	-0.215***	0.0044
HH Member Under 10	Response Rate	-0.044***	0.0040	-0.048***	0.0028
Any Non-Citizen in HH	Response Rate	-0.030***	0.0027	-0.022***	0.0016
Any Persons of Color in HH	Response Rate	-0.097***	0.0057	-0.074***	0.0034
HH Member Disabled	Response Rate	-0.028***	0.0027	-0.027***	0.0019
Married Couple in HH	Response Rate	-0.212***	0.0059	-0.181***	0.0040
Structure Type is a Single-Family Home/Townhouse	Response Rate	-0.126***	0.0061	-0.117***	0.0041
Income Over $130,000	Volunteering Rate	0.001	0.0103	-0.004	0.0071
Income Under $40,000	Volunteering Rate	0.000	0.0140	0.002	0.0091

HH Member Over 60	Volunteering Rate	0.003	0.0200	0.000	0.0131
HH Member Under 10	Volunteering Rate	-0.017	0.0123	-0.003	0.0085
Any Non-Citizen in HH	Volunteering Rate	-0.006	0.0083	-0.010*	0.0048
Any Persons of Color in HH	Volunteering Rate	-0.014	0.0175	-0.021*	0.0104
HH Member Disabled	Volunteering Rate	0.010	0.0082	0.003	0.0056
Married Couple in HH	Volunteering Rate	-0.016	0.0181	0.003	0.0121
Structure Type is a Single-Family Home/Townhouse	Volunteering Rate	-0.014	0.0185	-0.011	0.0124
Income Over $130,000	Voter Turnout	-0.004	0.0046	0.001	0.0029
Income Under $40,000	Voter Turnout	-0.004	0.0056	-0.001	0.0036
HH Member Over 60	Voter Turnout	-0.007	0.0090	-0.002	0.0053
HH Member Under 10	Voter Turnout	-0.012*	0.0052	0.000	0.0034
Any Non-Citizen in HH	Voter Turnout	-0.010**	0.0035	-0.004*	0.0019
Any Persons of Color in HH	Voter Turnout	0.007	0.0075	-0.001	0.0042
HH Member Disabled	Voter Turnout	-0.007*	0.0034	-0.001	0.0023
Married Couple in HH	Voter Turnout	-0.007	0.0077	0.004	0.0048
Structure Type is a Single-Family Home/Townhouse	Voter Turnout	-0.002	0.0079	-0.003	0.0051

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents spatial error regression model results of explanatory variables on bias estimates from the 2021 ACS sample year. Significance asterisks: *** p<.001 ** p<.01 * p<.05.

Table 9: Bias Regression Results for Spatial Lag Model

		CBSA		ACS Estimation Area	
Dependent Variable	Coefficient	Estimate	S.E.	Estimate	S.E
Income Over $130,000	Intercept	NA	NA	NA	NA
Income Under $40,000	Intercept	NA	NA	NA	NA
HH Member Over 60	Intercept	NA	NA	NA	NA
HH Member Under 10	Intercept	NA	NA	NA	NA
Any Non-Citizen in HH	Intercept	NA	NA	NA	NA
Any Persons of Color in HH	Intercept	NA	NA	NA	NA
HH Member Disabled	Intercept	NA	NA	NA	NA
Married Couple in HH	Intercept	NA	NA	NA	NA
Structure Type is a Single-Family Home/Townhouse	Intercept	NA	NA	NA	NA
Income Over $130,000	Civic Organization	-1.353	6.9550	8.204*	3.6040
Income Under $40,000	Civic Organization	3.010	10.1200	8.600	4.8870
HH Member Over 60	Civic Organization	9.774	12.7700	17.180*	6.9240
HH Member Under 10	Civic Organization	-4.156	8.4180	1.179	4.3780
Any Non-Citizen in HH	Civic Organization	-3.378	5.3680	0.458	2.6030
Any Persons of Color in HH	Civic Organization	-30.750**	11.7500	-8.653	5.6970

HH Member Disabled	Civic Organization	-8.659	5.7450	-2.870	2.9200
Married Couple in HH	Civic Organization	-11.550	12.2600	7.398	6.4600
Structure Type is a Single-Family Home/Townhouse	Civic Organization	1.226	11.5200	-3.943	6.2790
Income Over $130,000	Clustering	0.135***	0.0243	0.057***	0.0143
Income Under $40,000	Clustering	0.119***	0.0347	0.065**	0.0197
HH Member Over 60	Clustering	0.124**	0.0455	0.016	0.0273
HH Member Under 10	Clustering	0.115***	0.0288	0.000	0.0177
Any Non-Citizen in HH	Clustering	0.063**	0.0196	0.005	0.0101
Any Persons of Color in HH	Clustering	0.179***	0.0405	0.045*	0.0217
HH Member Disabled	Clustering	0.103***	0.0188	0.062***	0.0113
Married Couple in HH	Clustering	0.176***	0.0421	0.047	0.0254
Structure Type is a Single-Family Home/Townhouse	Clustering	0.113**	0.0408	0.067**	0.0252
Income Over $130,000	Economic Connectedness	-0.004	0.0057	0.000	0.0037
Income Under $40,000	Economic Connectedness	0.002	0.0091	-0.003	0.0053
HH Member Over 60	Economic Connectedness	0.033**	0.0108	0.002	0.0073
HH Member Under 10	Economic Connectedness	0.006	0.0077	0.002	0.0048
Any Non-Citizen in HH	Economic Connectedness	0.000	0.0049	-0.005	0.0028
Any Persons of Color in HH	Economic Connectedness	-0.016	0.0107	-0.006	0.0057
HH Member Disabled	Economic Connectedness	0.002	0.0051	0.002	0.0031
Married Couple in HH	Economic Connectedness	0.004	0.0115	-0.010	0.0069
Structure Type is a Single-Family Home/Townhouse	Economic Connectedness	0.025*	0.0109	0.005	0.0065
Income Over $130,000	Log Population Density	0.086**	0.0287	0.071***	0.0193
Income Under $40,000	Log Population Density	-0.064	0.0459	-0.082**	0.0264
HH Member Over 60	Log Population Density	0.154**	0.0560	0.119***	0.0354
HH Member Under 10	Log Population Density	-0.019	0.0345	-0.046	0.0242
Any Non-Citizen in HH	Log Population Density	-0.089***	0.0242	-0.068***	0.0136
Any Persons of Color in HH	Log Population Density	0.000	0.0481	0.014	0.0304

HH Member Disabled	Log Population Density	-0.067**	0.0247	-0.024	0.0149
Married Couple in HH	Log Population Density	0.077	0.0529	0.059	0.0343
Structure Type is a Single-Family Home/Townhouse	Log Population Density	0.184***	0.0504	0.093**	0.0335
Income Over $130,000	Percent Asian	0.006	0.0153	-0.005	0.0105
Income Under $40,000	Percent Asian	0.025	0.0238	0.031*	0.0137
HH Member Over 60	Percent Asian	-0.047	0.0300	0.008	0.0201
HH Member Under 10	Percent Asian	-0.014	0.0188	-0.022	0.0130
Any Non-Citizen in HH	Percent Asian	-0.054***	0.0117	-0.038***	0.0075
Any Persons of Color in HH	Percent Asian	-0.005	0.0265	-0.011	0.0153
HH Member Disabled	Percent Asian	0.000	0.0120	-0.002	0.0078
Married Couple in HH	Percent Asian	0.034	0.0270	0.037*	0.0187
Structure Type is a Single-Family Home/Townhouse	Percent Asian	0.022	0.0261	0.002	0.0173
Income Over $130,000	Percent Black	-0.002	0.0028	0.002	0.0017
Income Under $40,000	Percent Black	-0.003	0.0044	-0.001	0.0023
HH Member Over 60	Percent Black	0.003	0.0058	-0.002	0.0033
HH Member Under 10	Percent Black	-0.003	0.0038	0.004	0.0022
Any Non-Citizen in HH	Percent Black	-0.002	0.0025	-0.001	0.0013
Any Persons of Color in HH	Percent Black	-0.005	0.0052	0.004	0.0028
HH Member Disabled	Percent Black	0.003	0.0024	0.003	0.0014
Married Couple in HH	Percent Black	-0.019**	0.0057	-0.011**	0.0033
Structure Type is a Single-Family Home/Townhouse	Percent Black	-0.004	0.0056	-0.005	0.0031
Income Over $130,000	Percent Children	0.035**	0.0127	0.014	0.0082
Income Under $40,000	Percent Children	0.045*	0.0207	0.018	0.0121
HH Member Over 60	Percent Children	0.039	0.0255	-0.016	0.0159
HH Member Under 10	Percent Children	0.029	0.0161	0.030**	0.0108
Any Non-Citizen in HH	Percent Children	0.003	0.0110	0.018**	0.0059
Any Persons of Color in HH	Percent Children	-0.011	0.0218	0.028*	0.0136
HH Member Disabled	Percent Children	-0.009	0.0113	-0.005	0.0069
Married Couple in HH	Percent Children	0.022	0.0243	-0.018	0.0151
Structure Type is a Single-Family Home/Townhouse	Percent Children	0.024	0.0243	-0.014	0.0156
Income Over $130,000	Percent College	0.041***	0.0055	0.021***	0.0033
Income Under $40,000	Percent College	0.005	0.0078	-0.001	0.0046
HH Member Over 60	Percent College	-0.016	0.0102	-0.013*	0.0064
HH Member Under 10	Percent College	-0.001	0.0067	-0.009*	0.0041
Any Non-Citizen in HH	Percent College	0.002	0.0045	0.005*	0.0023
Any Persons of Color in HH	Percent College	0.015	0.0103	0.011*	0.0047
HH Member Disabled	Percent College	-0.004	0.0046	-0.011***	0.0026
Married Couple in HH	Percent College	0.028**	0.0102	0.011	0.0058

Structure Type is a Single-Family Home/Townhouse	Percent College	0.011	0.0098	0.018**	0.0058
Income Over $130,000	Percent Foreign Born	0.031***	0.0087	0.028***	0.0068
Income Under $40,000	Percent Foreign Born	-0.004	0.0144	-0.016	0.0094
HH Member Over 60	Percent Foreign Born	0.028	0.0180	-0.051***	0.0122
HH Member Under 10	Percent Foreign Born	0.003	0.0109	0.004	0.0081
Any Non-Citizen in HH	Percent Foreign Born	0.061***	0.0073	0.040***	0.0045
Any Persons of Color in HH	Percent Foreign Born	-0.019	0.0155	-0.027**	0.0098
HH Member Disabled	Percent Foreign Born	0.012	0.0075	0.004	0.0051
Married Couple in HH	Percent Foreign Born	-0.006	0.0163	-0.037**	0.0121
Structure Type is a Single-Family Home/Townhouse	Percent Foreign Born	0.016	0.0162	-0.004	0.0113
Income Over $130,000	Percent Hispanic	-0.010**	0.0031	-0.009***	0.0024
Income Under $40,000	Percent Hispanic	0.004	0.0050	0.003	0.0034
HH Member Over 60	Percent Hispanic	-0.014*	0.0063	0.007	0.0044
HH Member Under 10	Percent Hispanic	0.003	0.0036	-0.004	0.0029
Any Non-Citizen in HH	Percent Hispanic	-0.002	0.0026	-0.002	0.0017
Any Persons of Color in HH	Percent Hispanic	0.017**	0.0053	0.019***	0.0036
HH Member Disabled	Percent Hispanic	0.001	0.0027	-0.002	0.0019
Married Couple in HH	Percent Hispanic	-0.015**	0.0057	0.000	0.0044
Structure Type is a Single-Family Home/Townhouse	Percent Hispanic	-0.016**	0.0058	-0.009*	0.0044
Income Over $130,000	Percent Over 65	-0.014	0.0097	-0.019**	0.0073
Income Under $40,000	Percent Over 65	0.015	0.0158	-0.009	0.0105
HH Member Over 60	Percent Over 65	0.077***	0.0189	0.032*	0.0142
HH Member Under 10	Percent Over 65	0.019	0.0125	0.001	0.0091
Any Non-Citizen in HH	Percent Over 65	-0.012	0.0085	-0.004	0.0051
Any Persons of Color in HH	Percent Over 65	-0.033	0.0174	0.005	0.0116
HH Member Disabled	Percent Over 65	0.006	0.0087	0.001	0.0057
Married Couple in HH	Percent Over 65	-0.012	0.0184	-0.037**	0.0128
Structure Type is a Single-Family Home/Townhouse	Percent Over 65	-0.032	0.0189	-0.035**	0.0130
Income Over $130,000	Percent Poverty	-0.038***	0.0071	-0.035***	0.0051
Income Under $40,000	Percent Poverty	-0.003	0.0110	-0.009	0.0071
HH Member Over 60	Percent Poverty	-0.028*	0.0132	-0.028**	0.0099
HH Member Under 10	Percent Poverty	-0.007	0.0095	-0.003	0.0063

Any Non-Citizen in HH	Percent Poverty	-0.026***	0.0059	-0.018***	0.0037
Any Persons of Color in HH	Percent Poverty	-0.031*	0.0126	-0.012	0.0075
HH Member Disabled	Percent Poverty	-0.005	0.0062	0.004	0.0041
Married Couple in HH	Percent Poverty	-0.046***	0.0133	-0.048***	0.0094
Structure Type is a Single-Family Home/Townhouse	Percent Poverty	0.013	0.0131	-0.003	0.0098
Income Over $130,000	Response Rate	-0.090***	0.0033	-0.079***	0.0024
Income Under $40,000	Response Rate	-0.066***	0.0047	-0.060***	0.0032
HH Member Over 60	Response Rate	-0.280***	0.0070	-0.224***	0.0047
HH Member Under 10	Response Rate	-0.043***	0.0040	-0.048***	0.0030
Any Non-Citizen in HH	Response Rate	-0.031***	0.0027	-0.021***	0.0017
Any Persons of Color in HH	Response Rate	-0.101***	0.0056	-0.073***	0.0035
HH Member Disabled	Response Rate	-0.027***	0.0028	-0.029***	0.0018
Married Couple in HH	Response Rate	-0.218***	0.0066	-0.181***	0.0045
Structure Type is a Single-Family Home/Townhouse	Response Rate	-0.136***	0.0061	-0.124***	0.0044
Income Over $130,000	Volunteering Rate	0.004	0.0097	-0.004	0.0068
Income Under $40,000	Volunteering Rate	-0.001	0.0145	0.002	0.0087
HH Member Over 60	Volunteering Rate	0.011	0.0192	0.001	0.0127
HH Member Under 10	Volunteering Rate	-0.016	0.0130	-0.003	0.0083
Any Non-Citizen in HH	Volunteering Rate	-0.006	0.0085	-0.010*	0.0046
Any Persons of Color in HH	Volunteering Rate	-0.016	0.0165	-0.022*	0.0109
HH Member Disabled	Volunteering Rate	0.010	0.0083	0.003	0.0055
Married Couple in HH	Volunteering Rate	-0.015	0.0175	0.003	0.0122
Structure Type is a Single-Family Home/Townhouse	Volunteering Rate	-0.016	0.0183	-0.015	0.0124
Income Over $130,000	Voter Turnout	-0.006	0.0040	0.000	0.0026
Income Under $40,000	Voter Turnout	-0.005	0.0064	-0.001	0.0037
HH Member Over 60	Voter Turnout	-0.004	0.0078	0.000	0.0049
HH Member Under 10	Voter Turnout	-0.011*	0.0051	0.000	0.0034
Any Non-Citizen in HH	Voter Turnout	-0.010**	0.0035	-0.004*	0.0019
Any Persons of Color in HH	Voter Turnout	0.009	0.0072	0.000	0.0041
HH Member Disabled	Voter Turnout	-0.007*	0.0034	-0.001	0.0021
Married Couple in HH	Voter Turnout	-0.006	0.0074	0.004	0.0049
Structure Type is a Single-Family Home/Townhouse	Voter Turnout	-0.002	0.0073	-0.003	0.0049

Source: 2021 ACS Sample Year, 2015-2019 ACS 5-year, Administrative Data, Facebook data from Chetty et al. (2022), and voter turnout data from MIT Election Data and Science Lab (2018). Table presents spatial error regression model results of explanatory variables on bias estimates from the 2021 ACS sample year. Displayed coefficients are the marginal effects. Significance asterisks: *** p<.001 ** p<.01 * p<.05.